Optimization TA 4

Kacper Krasowski

Exercise 4.4

Problem

Consider a function $f : X \to Y$, where Y is an arbitrary metric space and $\langle X, \rho \rangle$ is endowed with the discrete metric. Show that every such function is continuous on X.

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$$\boxed{\langle X, \rho_d \rangle \text{ and arbitrary } \langle Y, \rho_Y \rangle \Rightarrow f : X \to Y \text{ Continuous}}$$

Exercise 4.4: Definitions

$$\boxed{\langle X, \rho_d \rangle \text{ and arbitrary } \langle Y, \rho_Y \rangle \Rightarrow f : X \to Y \text{ Continuous}}$$

Definitions

Discrete metric: A metric ρ on X is discrete if

$$\rho(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

$\epsilon - \delta$ **continuity:** A function $f : X \to Y$ is continuous at x_0 if, for every $\epsilon > 0$, there exists $\delta_\epsilon > 0$ such that

$$\rho_X(x, x_0) < \delta_\epsilon \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon.$$

Continuity: A function $f : X \to Y$ is continuous if it is continuous at every point $x_0 \in X$.

Exercise 4.4: Sketch of the Proof

$$\boxed{\langle X, \rho_d \rangle \text{ and arbitrary } \langle Y, \rho_Y \rangle \Rightarrow f : X \to Y \text{ Continuous}}$$

1. Apply definitions.
2. Set δ_ϵ such that $\rho_Y(f(x), f(x_0)) < \epsilon$ is always true.
3. Verify.

Exercise 4.5

Problem

Consider the functional $f : \mathbb{R} \to \mathbb{R}$ with

$$f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Z} \text{ is an integer,} \\ 0, & \text{otherwise.} \end{cases}$$

Identify the points at which f is continuous.

Exercise 4.5

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$$f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Z} \text{ is an integer,} \\ 0, & \text{otherwise.} \end{cases}$$

Identify the points at which f is continuous.

Find points $x_0 \in \mathbb{R}$ at which f is Continuous

Exercise 4.5: Definitions

Find points $x_0 \in \mathbb{R}$ at which f is Continuous

Definitions

Real numbers: $\mathbb{R}$ is the set of all real numbers.

Integers: $\mathbb{Z}$ is the set of all integers,

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

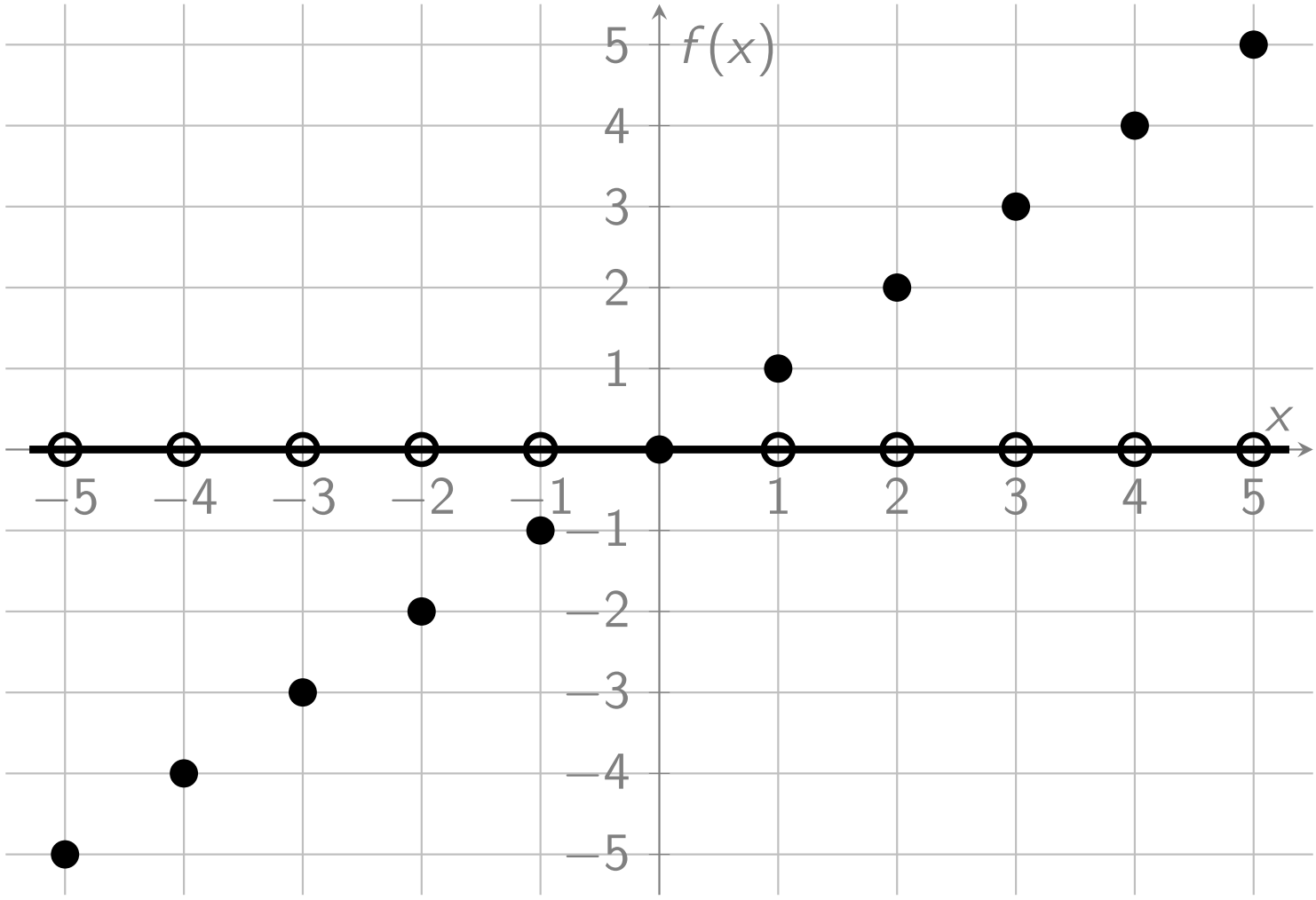
ϵ-δ **continuity:** A function $f : X \to Y$ is continuous at x_0 if, for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\rho_X(x, x_0) < \delta \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon.$$

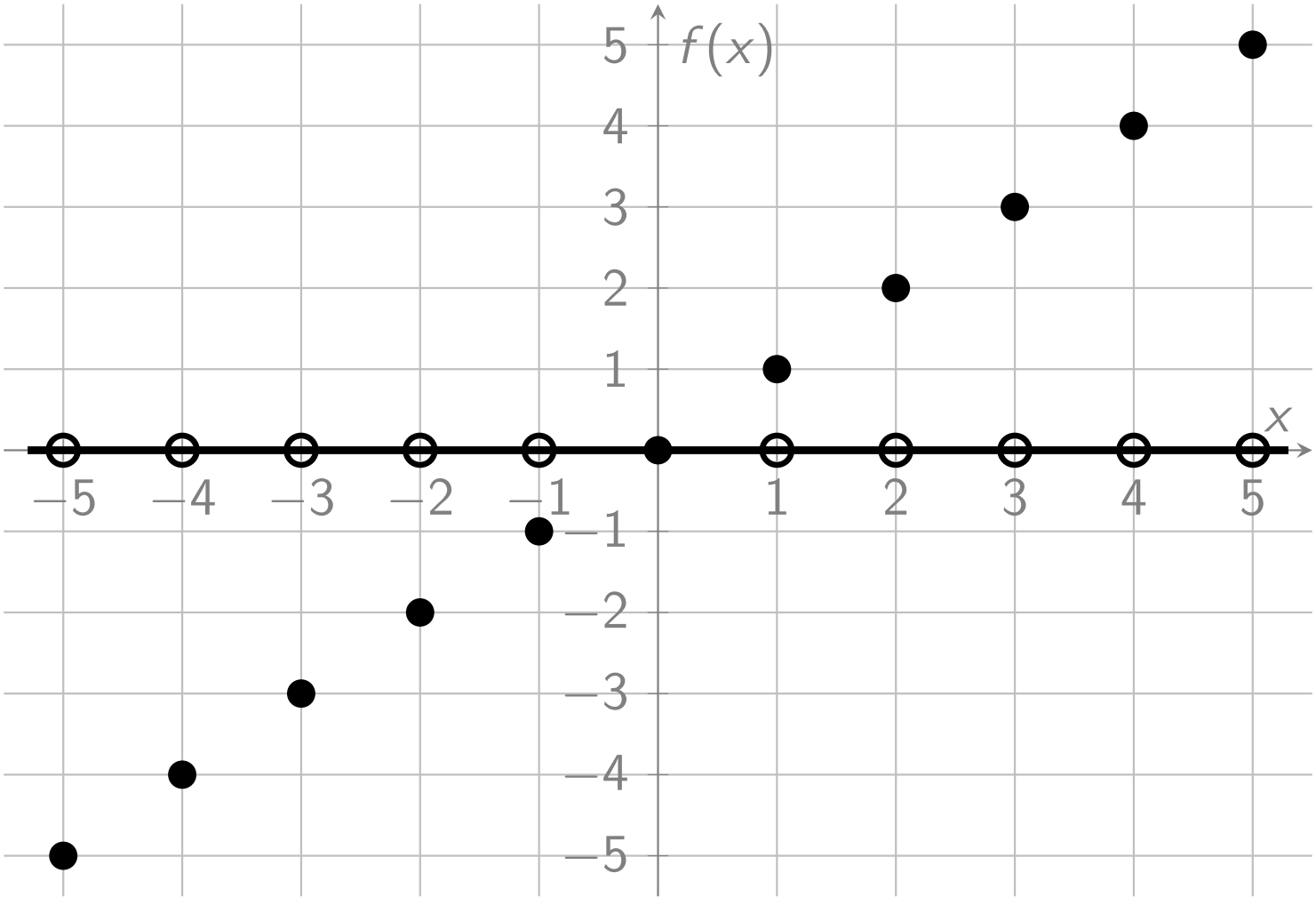
Reminder: On $\mathbb{R}$, the standard metric is

$$\rho_2(x, x_0) = |x - x_0|.$$

Exercise 4.5: Intuition

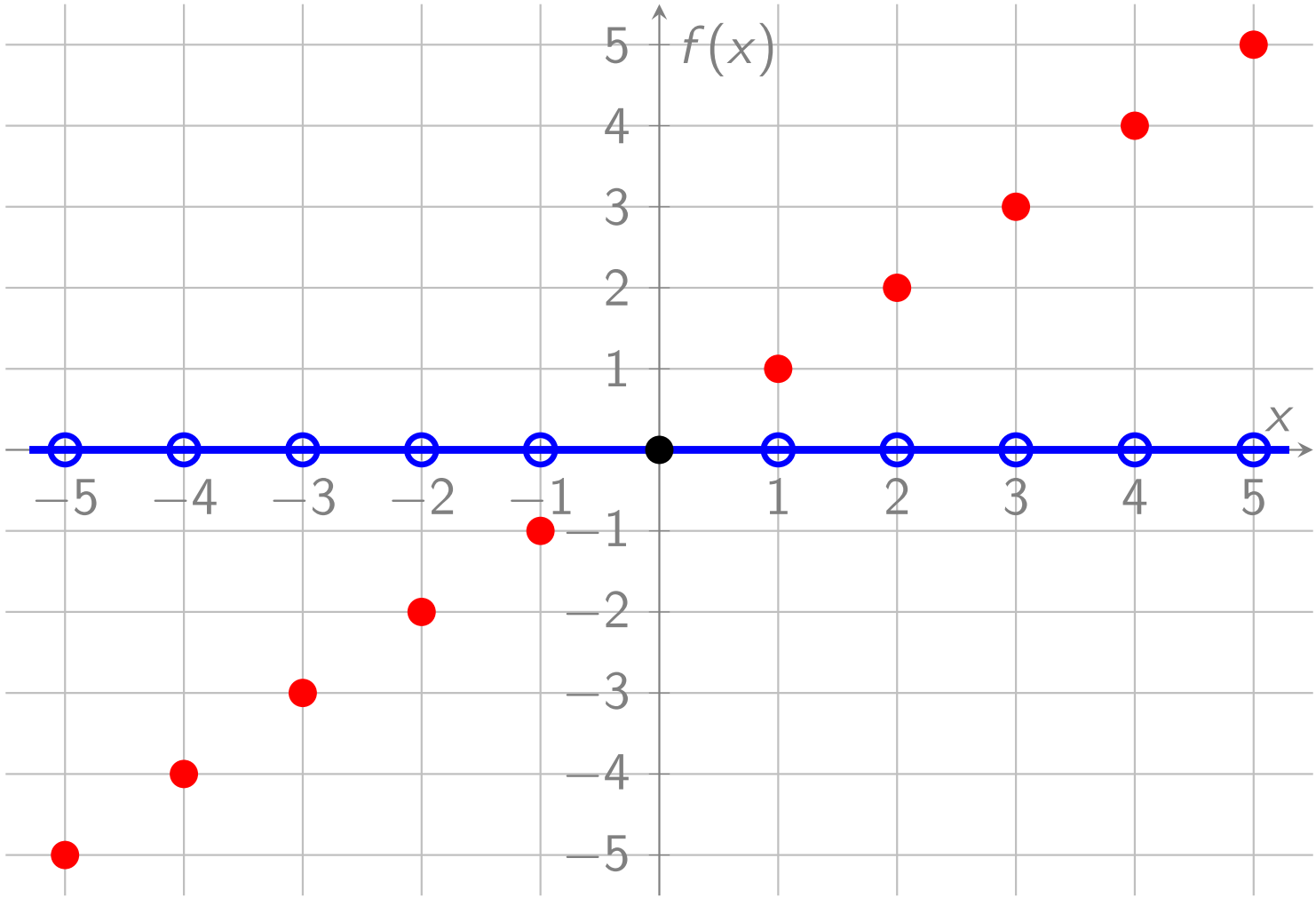


Exercise 4.5: Intuition



Where do we expect the Continuity/Discontinuity?

Exercise 4.5: Intuition



Where do we expect the Continuity/Discontinuity?

Continuity: $\mathbb{R}/\mathbb{Z}$, **Discontinuity**: $\mathbb{Z}$

Exercise 4.5: Sketch of the Proof

Find points $x_0 \in \mathbb{R}$ at which f is Continuous

1. Prove Continuity at $\mathbb{R}/\mathbb{Z}$, by finding $\delta_\epsilon(\epsilon, x_0)$, that satisfies continuity.
2. Disprove Continuity at $\mathbb{Z}$, by finding ϵ_0, such that all $\delta_{\epsilon_0} > 0$ do not satisfy continuity.
3. End?

▸ Appendix: Disproving ▸ Appendix: Negating continuity

Exercise 4.6

Problem

Show that $f : X \to Y$ is continuous if and only if $f(x) = \lim_{n \to +\infty} f(x^n)$ for all sequences $x^n \to x$ converging to x.

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$$\boxed{f \text{ Continuous} \iff (x^n \to x \Rightarrow f(x^n) \to f(x))}$$

This can be split into:

Exercise 4.6

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Show that $f : X \to Y$ is continuous if and only if $f(x) = \lim_{n \to +\infty} f(x^n)$ for all sequences $x^n \to x$ converging to x.

$$\boxed{f \text{ Continuous } \iff (x^n \to x \Rightarrow f(x^n) \to f(x))}$$

This can be split into:

$$\boxed{f \text{ continuous on } X \text{ and } x^n \to x \Longrightarrow f(x^n) \to f(x)}$$

Exercise 4.6

Problem

Show that $f : X \to Y$ is continuous if and only if $f(x) = \lim_{n \to +\infty} f(x^n)$ for all sequences $x^n \to x$ converging to x.

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$$\boxed{f \text{ continuous on } X \text{ and } x^n \to x \Longrightarrow f(x^n) \to f(x)}$$

$$\boxed{(x^n \to x \Rightarrow f(x^n) \to f(x)) \Longrightarrow f \text{ continuous on } X}$$

Exercise 4.6: First Direction

$$\boxed{f \text{ continuous on } X \text{ and } x^n \to x \Longrightarrow f(x^n) \to f(x)}$$

1. Take any arbitrary ϵ, and use the definition of continuity to get δ_ϵ.
2. By convergence, get N for δ_ϵ (or A).
3. Apply continuity for the same N (or A) to verify convergence of $f(x^n)$.

▸ Appendix: Definition of convergence

Exercise 4.6: Second Direction

$$\boxed{(x^n \to x \Rightarrow f(x^n) \to f(x)) \Longrightarrow f \text{ continuous on } X}$$

1. Contrapositive.

▸ Appendix: Negating continuity ▸ Appendix: Constructing the sequence

Exercise 4.6: Second Direction

$$f \text{ not continuous at } x \implies \text{There exists } (x^n) \to x : f(x^n) \not\to f(x)$$

1. Contrapositive.

▸ Appendix: Negating continuity ▸ Appendix: Constructing the sequence

Exercise 4.6: Second Direction

$$f \text{ not continuous at } x \implies \text{There exists } (x^n) \to x : f(x^n) \not\to f(x)$$

1. Contrapositive.
2. Negate continuity.
3. Define δ^n to be such that x^n converges.
4. Use the definition of negated continuity to show that $f(x^n)$ does not converge to $f(x)$.

▸ Appendix: Negating continuity ▸ Appendix: Constructing the sequence

Exercise 4.7

Problem

Show that graph(f) is a closed subset of $X \times Y$ for every continuous function $f : X \to Y$.

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$$\boxed{f : X \to Y \text{ continuous} \Longrightarrow \text{graph}(f) \text{ is closed in } X \times Y}$$

Exercise 4.7: Definitions

$$\boxed{f : X \to Y \text{ continuous} \Longrightarrow \text{graph}(f) \text{ is closed in } X \times Y}$$

Definitions

Continuous: A function $f : X \to Y$ is continuous if, for every sequence $x^n \to x$ in X, $f(x^n) \to f(x)$ in Y. (Ex4.6)

Closed: A set S is closed if, for every sequence $(x^n) \subseteq S$ such that $x^n \to x$, we have $x \in S$.

Graph of a function: The graph of $f : X \to Y$ is the set

$$\text{graph}(f) = \{(x, y) \in X \times Y : y = f(x) \in X\}.$$

Exercise 4.7: Sketch of Proof

$$\boxed{f : X \to Y \text{ continuous} \Longrightarrow \text{graph}(f) \text{ is closed in } X \times Y}$$

1. Take any arbitrary $(x^n, y^n) \in \text{graph}(f) : \ (x^n, y^n) \to (x, y)$.
2. Establish that $y^n = f(x^n) \to f(x)$.
3. Show that $(x, f(x)) \in \text{graph}(f)$.

Exercise 5.1

Problem

Intermediate Value Theorem. Let $f : [a, b] \to \mathbb{R}$ be continuous on the closed interval $[a, b]$ with $f(a) < f(b)$. Show that for every $y \in (f(a), f(b))$ there exists some $c \in (a, b)$ such that $f(c) = y$.

Exercise 5.1

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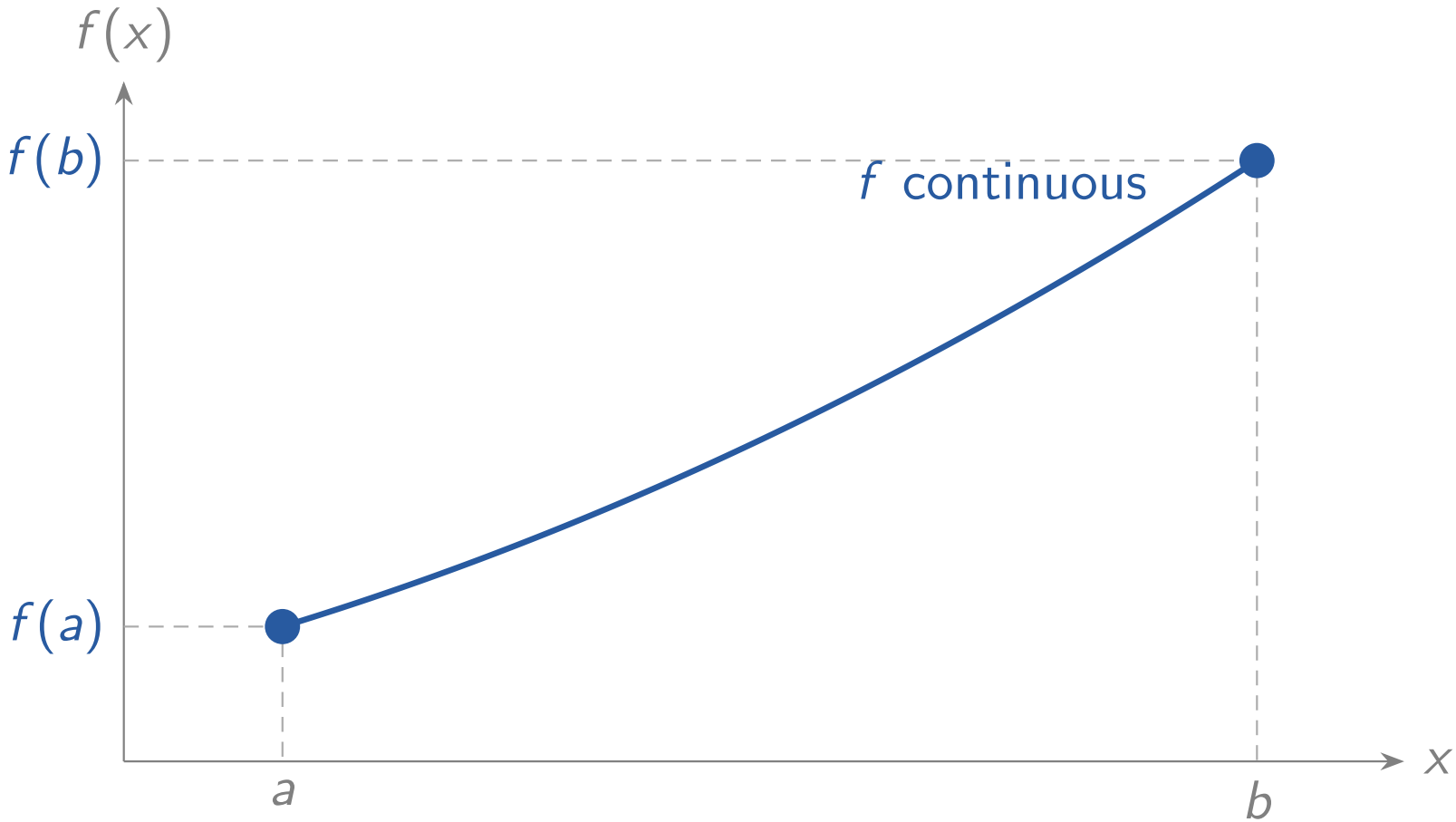
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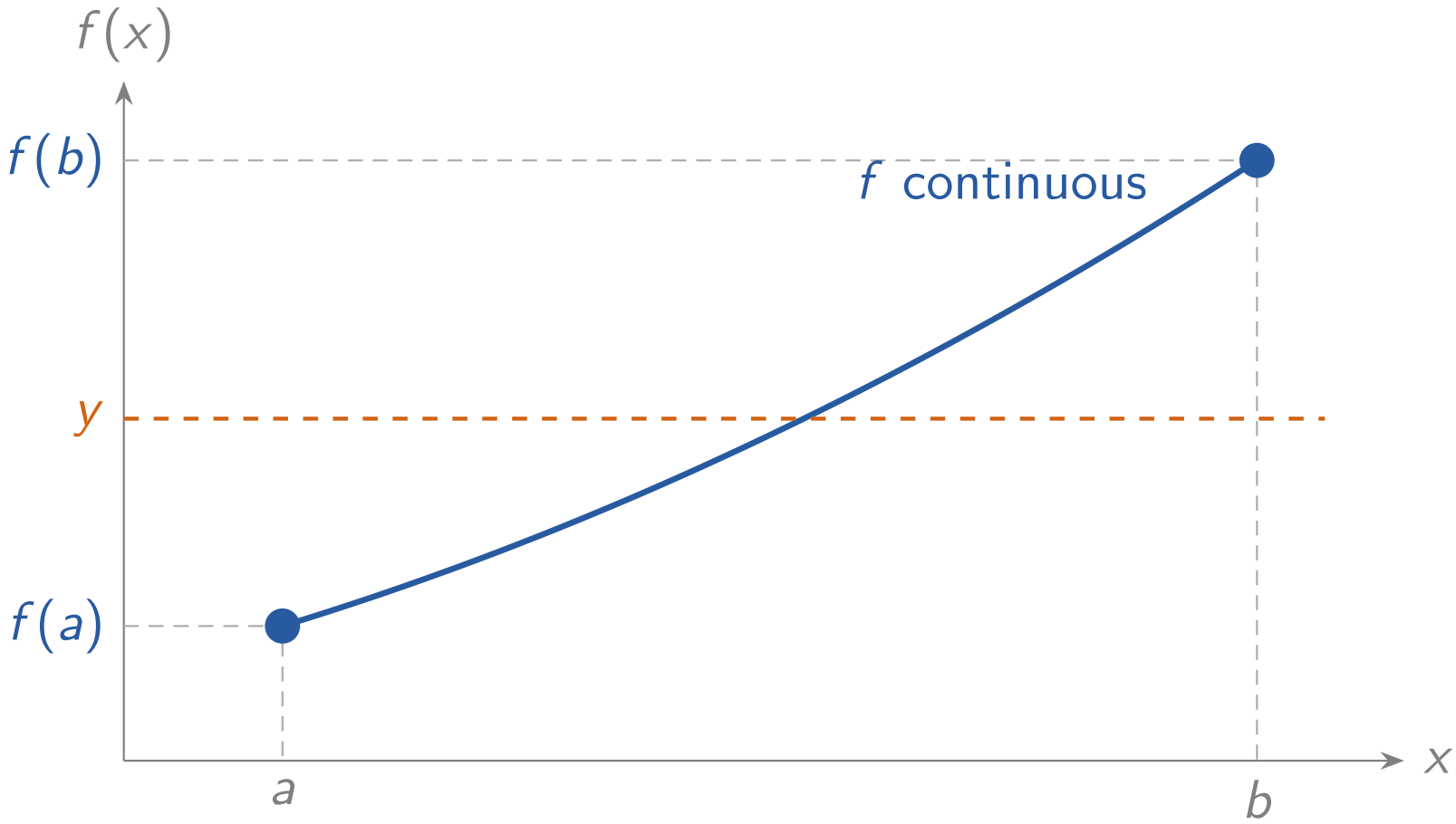
$$f : [a, b] \to \mathbb{R} \text{ continuous}, y \in (f(a), f(b)) \Longrightarrow c \in (a, b) : f(c) = y$$

Exercise 5.1: Intuition



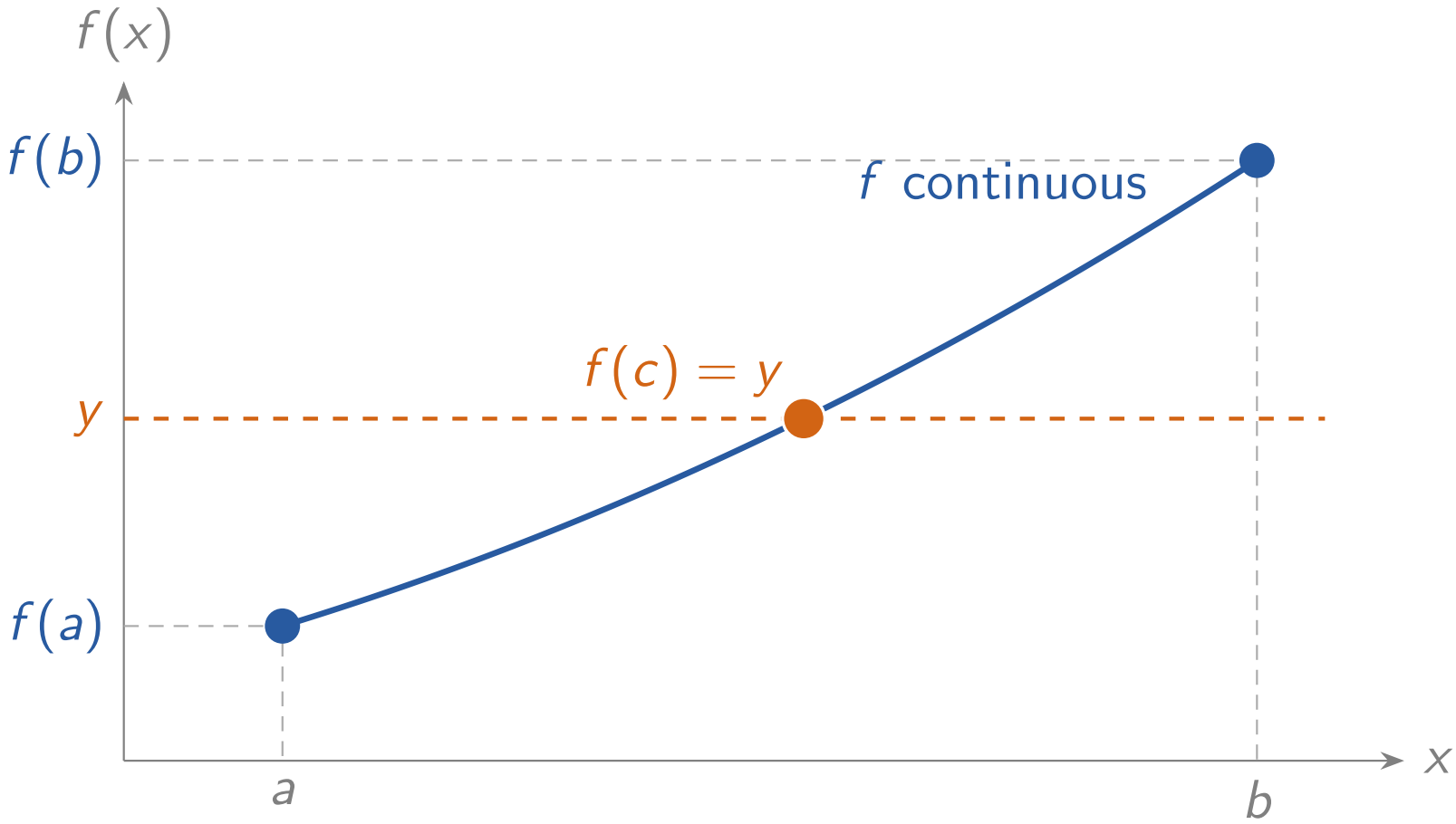
A continuous curve joins the endpoint values $f(a)$ and $f(b)$.

Exercise 5.1: Intuition



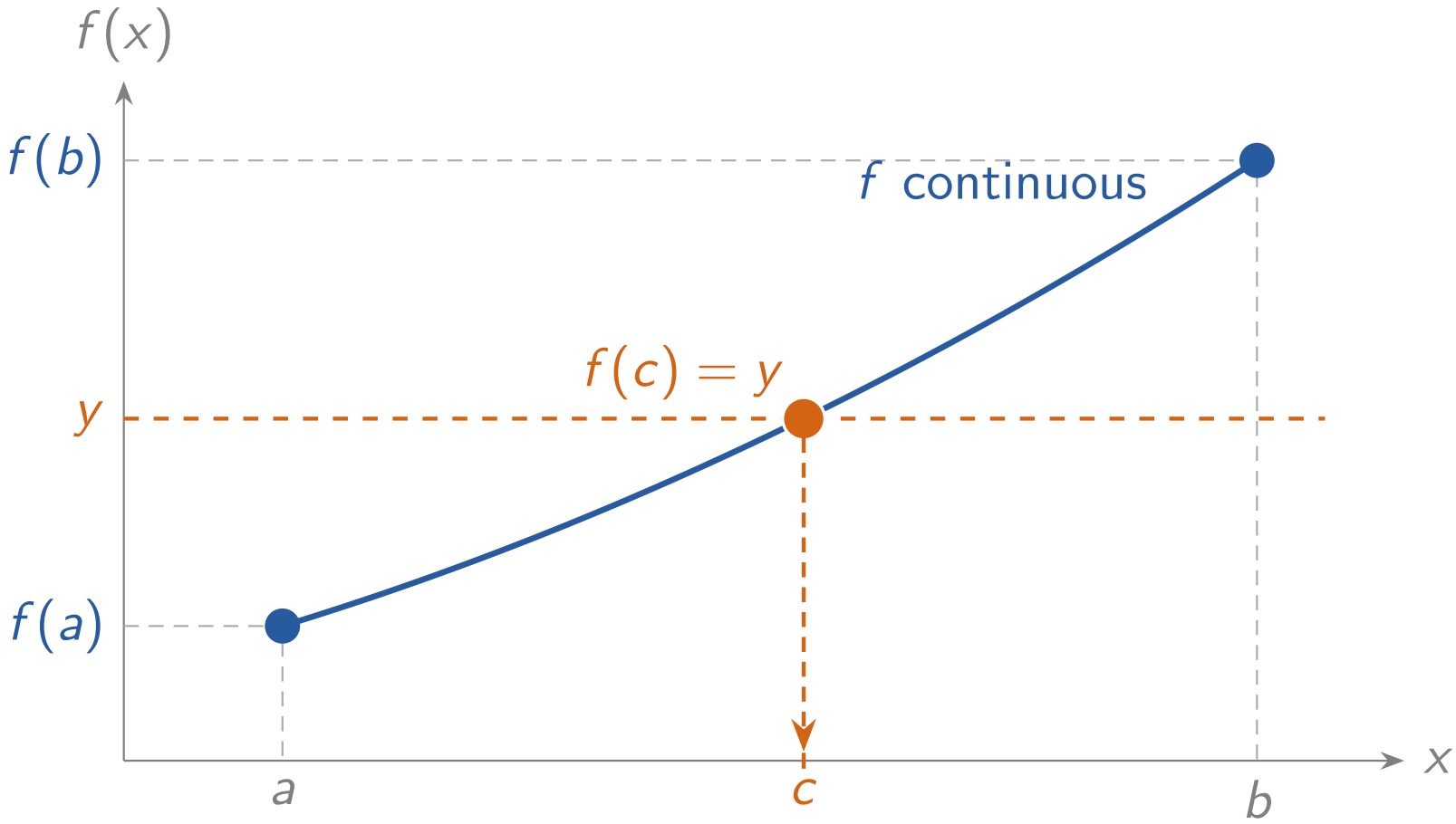
Choose any height $y \in (f(a), f(b))$.

Exercise 5.1: Intuition



The curve must cross this height: $f(c) = y$.

Exercise 5.1: Intuition



There exists $c \in (a, b)$ such that $f(c) = y$

Exercise 5.1: Definitions

$$f : [a, b] \to \mathbb{R} \text{ continuous}, y \in (f(a), f(b)) \implies c \in (a, b) : f(c) = y$$

Definitions

Supremum: The supremum of a set $S \subseteq \mathbb{R}$ is its smallest upper bound. That is, $c = \sup S$ if

$$x \leq c \quad \text{for all } x \in S,$$

and for every $\epsilon > 0$ there exists $x \in S$ such that

$$c - \epsilon < x \leq c.$$

Continuous: A function $f : [a, b] \to \mathbb{R}$ is continuous if for all $x_0 \in [a, b]$, for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$|x - x_0| < \delta \quad \implies \quad |f(x) - f(x_0)| < \epsilon.$$

Exercise 5.1: Sketch of the Proof

$$f : [a, b] \to \mathbb{R} \text{ continuous}, y \in (f(a), f(b)) \Longrightarrow c \in (a, b) : f(c) = y$$

1. Define $L = \{x \in [a, b] : f(x) < y\}$. Show by contradiction, that $c = \sup L \notin L$.
2. Since $c \notin L$, then $f(c) \geq y$
3. Similarly to 1., show that $f(c) > y$ leads to a contradiction.
4. Conclude that $f(c) = y$, as $c \notin L$, and $f(c) > y$ leads to a contradiction

Exercise 5.2

Problem

Show that a functional $f : X \to \mathbb{R}$ is continuous if and only if it is both upper semicontinuous and lower semicontinuous.

f Continous $\Leftrightarrow$ f Upper Semicontinuous and Lower Semicontinuous

Exercise 5.2

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f Continous $\Leftrightarrow$ f Upper Semicontinuous and Lower Semicontinuous

Exercise 5.2: Definitions

f Continous $\Leftrightarrow$ f Upper Semicontinuous and Lower Semicontinuous

Definitions

Preimage: For a set $A \subseteq \mathbb{R}$, $f^{-1}(A) = \{x \in X : f(x) \in A\}$.

Continuous (closed(open)-set characterization): A function $f : X \to \mathbb{R}$ is continuous if, for every closed(open) set $C \subseteq \mathbb{R}$, $f^{-1}(C)$ is closed(open) in X.

Upper semicontinuous (usc): A function $f : X \to \mathbb{R}$ is upper semicontinuous if, for every $\alpha \in \mathbb{R}$, $\{x \in X : f(x) \geq \alpha\}$ is closed in X.

Lower semicontinuous (lsc): A function $f : X \to \mathbb{R}$ is lower semicontinuous if, for every $\alpha \in \mathbb{R}$, $\{x \in X : f(x) \leq \alpha\}$ is closed in X.

▸ Appendix: Open and closed sets

Exercise 5.2: First Direction

f Continous $\Rightarrow$ f Upper Semicontinuous and Lower Semicontinuous

1. Notice that $[\alpha, \infty)$ is closed and its preimage is exactly the $\{x \in X : f(x) \geq \alpha\}$.
2. Using Continuity, we can say that $\{x \in X : f(x) \geq \alpha\}$ is closed, which gives us the usc.
3. The argument for lsc is symmetrical.

▸ Appendix: Closed-set characterization

Exercise 5.2: Second Direction

f Upper Semicontinuous and Lower Semicontinuous $\Rightarrow$ f Continous

1. Notice that $\{x \in X : f(x) \geq a+\epsilon\}^c = \{x \in X : f(x) < a+\epsilon\}$ is open by usc, same with $\{x \in X : f(x) > a-\epsilon\}$ by lsc.
2. Take any arbitrary open interval $(a-\epsilon, a+\epsilon)$. Notice that $f^{-1}((a-\epsilon, a+\epsilon)) = \{x \in X : f(x) > a-\epsilon\} \cap \{x \in X : f(x) < a+\epsilon\}$.
3. This is exactly the open-set characterization of continuity.

▸ Appendix: Open-set characterization

Exercise 5.3

Problem

Now consider a continuous functional on the space of bounded sequences. Denote by ℓ^∞ the space of bounded real sequences, formally

$$\ell^\infty = \left\{ (x^n) \in \mathbb{R}^\infty : \sup_{n\in\mathbb{N}} |x^n| < \infty \right\}.$$

To make this a metric space, endow ℓ^∞ with the sup metric $\rho_\infty\big((x^n),(y^n)\big) = \sup_{n\in\mathbb{N}} |x^n - y^n|$. Now define a functional $f : \ell^\infty \to \mathbb{R}_+$
by

$$f\big((x^n)\big) = \sup_{n\in\mathbb{N}} |x^n|.$$

Show that f is continuous on this metric space.

$$\boxed{\langle \ell^\infty, \rho_\infty \rangle, \langle \mathbb{R}_+, \rho_2 \rangle, f\big((x^n)\big) = \sup |x^n| \Rightarrow f \text{ Continous}}$$

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$$\boxed{\langle \ell^\infty, \rho_\infty \rangle, \langle \mathbb{R}_+, \rho_2 \rangle, f\big((x^n)\big) = \sup |x^n| \Rightarrow f \text{ Continous}}$$

Exercise 5.3: Definitions

$$\boxed{\langle \ell^\infty, \rho_\infty \rangle, \langle \mathbb{R}_+, \rho_2 \rangle, f\left((x^n)\right) = \sup |x^n| \Rightarrow f \text{ Continous}}$$

Definitions

Continuous: A function $f : X \to Y$ is continuous at $x_0 \in X$ if, for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$\rho_X(x, x_0) < \delta \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon.$$

The function f is continuous on X if it is continuous at every $x_0 \in X$.

Exercise 5.3: Sketch of the Proof

$$\boxed{\langle \ell^\infty, \rho_\infty \rangle, \langle \mathbb{R}_+, \rho_2 \rangle, f\left((x^n)\right) = \sup |x^n| \Rightarrow f \text{ Continous}}$$

1. Apply $\langle \ell^\infty, \rho_\infty \rangle, \langle \mathbb{R}_+, \rho_2 \rangle, f\left((x^n)\right) = \sup |x^n|$ to the definition of continuity.
2. Find δ such that the definition is true.

Exercise 5.4

Problem

Consider the square root function $f : [0,1] \to [0,1], f(x) = \sqrt{x}$, under the standard metric. Check whether f is uniformly continuous and whether it is Lipschitz continuous on $[0,1]$.

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Consider the square root function $f : [0,1] \to [0,1]$, $f(x) = \sqrt{x}$, under the standard metric. Check whether f is uniformly continuous and whether it is Lipschitz continuous on $[0,1]$.

$f(x) = \sqrt{x}$ on $[0,1]$: Uniformly continuous? Lipschitz?

Exercise 5.4: Definitions

Definitions

Uniformly continuous: A function $f : X \to Y$ is uniformly continuous if, for every $\epsilon > 0$, there exists $\delta_\epsilon > 0$ such that, for all $x, y \in X$,

$$\rho_X(x, y) < \delta_\epsilon \implies \rho_Y(f(x), f(y)) < \epsilon.$$

The same δ_ϵ must work for all $x, y \in X$.

Lipschitz continuous: A function $f : X \to Y$ is Lipschitz continuous if there exists a finite constant $L \geq 0$ such that, for all $x, y \in X$,

$$\rho_Y(f(x), f(y)) \leq L\rho_X(x, y).$$

One finite constant L must work for all pairs.

Here, $\rho_X(x, y) = |x - y|$ and $\rho_Y(f(x), f(y)) = |\sqrt{x} - \sqrt{y}|$.

Exercise 5.4: Uniform Continuity

$$\boxed{f(x) = \sqrt{x} \text{ is uniformly continuous on } [0, 1]}$$

1. Take any arbitrary $\epsilon > 0$. We need one δ_ϵ that works for every $x, y \in [0, 1]$.

Exercise 5.4: Uniform Continuity

$$\boxed{f(x) = \sqrt{x} \text{ is uniformly continuous on } [0, 1]}$$

1. Take any arbitrary $\epsilon > 0$. We need one δ_ϵ that works for every $x, y \in [0, 1]$.
2. Suppose $x \geq y$. Notice that

$$(\sqrt{x} - \sqrt{y})^2 \leq (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) = x - y.$$

   By symmetry, $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$.

Exercise 5.4: Uniform Continuity

$$\boxed{f(x) = \sqrt{x} \text{ is uniformly continuous on } [0, 1]}$$

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   By symmetry, $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$.
3. Set $\delta_\epsilon = \epsilon^2$. Verify:

$$|x - y| < \delta_\epsilon \implies |\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|} < \epsilon.$$

Exercise 5.4: Lipschitz Continuity

$$\boxed{f(x) = \sqrt{x} \text{ is not Lipschitz continuous on } [0, 1]}$$

1. Negate the definition. For every finite constant $L \geq 0$, we need to find points $x, y \in [0, 1]$ such that

$$|\sqrt{x} - \sqrt{y}| > L|x - y|.$$

▸ Appendix: Disproving

Exercise 5.4: Lipschitz Continuity

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1. Negate the definition. For every finite constant $L \geq 0$, we need to find points $x, y \in [0, 1]$ such that

$$|\sqrt{x} - \sqrt{y}| > L|x - y|.$$

2. Take any arbitrary $L \geq 0$ and set $y = 0$. For $x > 0$, the required inequality becomes

$$\frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} > L.$$

▸ Appendix: Disproving

Exercise 5.4: Lipschitz Continuity

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1. Negate the definition. For every finite constant $L \geq 0$, we need to find points $x, y \in [0, 1]$ such that

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2. Take any arbitrary $L \geq 0$ and set $y = 0$. For $x > 0$, the required inequality becomes

$$\frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} > L.$$

3. Choose $x = (L + 1)^{-2} \in (0, 1]$. Then $1/\sqrt{x} = L + 1 > L$. No finite L works for all pairs.

▸ Appendix: Disproving

Appendix: Disproving a Statement

To disprove a statement, prove its negation.

1. To disprove *Every object has a certain property*, find **at least one** object without that property.
   For example, to disprove *Every integer is positive*, take the integer 0.

▸ Apply to continuity ▸ Cheat sheet ◂ Exercise 4.5 ◂ Exercise 5.4

Appendix: Disproving a Statement

To disprove a statement, prove its negation.

1. To disprove *Every object has a certain property*, find **at least one** object without that property.
   For example, to disprove *Every integer is positive*, take the integer 0.
2. To disprove *There exists an object with a certain property*, show that **every** object fails to have it.

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1. To disprove *Every object has a certain property*, find **at least one** object without that property.
   For example, to disprove *Every integer is positive*, take the integer 0.
2. To disprove *There exists an object with a certain property*, show that **every** object fails to have it.
3. To disprove *If an assumption holds, then a conclusion follows*, find a case where the assumption is true **and** the conclusion is false.

▸ Apply to continuity ▸ Cheat sheet ◂ Exercise 4.5 ◂ Exercise 5.4

Appendix: Disproving Continuity

Not continuous at x_0

There exists $\epsilon_0 > 0$ such that, for every $\delta > 0$, there exists a point $x \in X$ with

$$\rho_X(x, x_0) < \delta \quad \text{and} \quad \rho_Y(f(x), f(x_0)) \geq \epsilon_0.$$

1. Fix one $\epsilon_0 > 0$: a gap in function values that cannot be avoided near x_0.

Failing to find a working δ is not itself a disproof.

▸ Derive the definition ▸ Cheat sheet ◂ Exercise 4.5

Appendix: Disproving Continuity

Not continuous at x_0

There exists $\epsilon_0 > 0$ such that, for every $\delta > 0$, there exists a point $x \in X$ with

$$\rho_X(x, x_0) < \delta \quad \text{and} \quad \rho_Y(f(x), f(x_0)) \geq \epsilon_0.$$

1. Fix one $\epsilon_0 > 0$: a gap in function values that cannot be avoided near x_0.
2. Take any arbitrary $\delta > 0$. Find a point $x \in X$ satisfying both inequalities. The point may depend on δ.

Failing to find a working δ is not itself a disproof.

▸ Derive the definition ▸ Cheat sheet ◂ Exercise 4.5

Appendix: Disproving Continuity

Not continuous at x_0

There exists $\epsilon_0 > 0$ such that, for every $\delta > 0$, there exists a point $x \in X$ with

$$\rho_X(x, x_0) < \delta \quad \text{and} \quad \rho_Y(f(x), f(x_0)) \geq \epsilon_0.$$

1. Fix one $\epsilon_0 > 0$: a gap in function values that cannot be avoided near x_0.
2. Take any arbitrary $\delta > 0$. Find a point $x \in X$ satisfying both inequalities. The point may depend on δ.
3. Verify that your construction works for every $\delta > 0$, with the same ϵ_0.

Failing to find a working δ is not itself a disproof.

▸ Derive the definition ▸ Cheat sheet ◂ Exercise 4.5

Appendix: Disproving — Cheat Sheet

Statement	**Its negation**
For every $x \in X$, $P(x)$ is true.	There exists $x \in X$ for which $P(x)$ is false.
There exists $x \in X$ for which $P(x)$ is true.	For every $x \in X$, $P(x)$ is false.
$a < b$	$a \geq b$
$a \leq b$	$a > b$
$a > b$	$a \leq b$
$a \geq b$	$a < b$
$a = b$	$a \neq b$
$a \neq b$	$a = b$
A and B are true.	A is false or B is false.
A or B is true.	A is false and B is false.
If A is true, then B is true.	A is true and B is false.

$P(x), A, B$ stand for statements; a, b are real numbers. Here, *or* includes the case where both are true.

Keep the domain and order of choices. *For every* $\epsilon > 0$ becomes *there exists* $\epsilon > 0$; keep $\epsilon > 0$.

Appendix: Negating Continuity Step by Step

Definition

A function $f : X \to Y$ is continuous at x_0 if, for every $\epsilon > 0$, there exists $\delta > 0$ such that, for every $x \in X$,

$$\rho_X(x, x_0) < \delta \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon.$$

Appendix: Negating Continuity Step by Step

Definition

A function $f : X \to Y$ is **continuous** at x_0 if, **for every $\epsilon > 0$**, **there exists $\delta > 0$** such that, **for every $x \in X$**,

$$\rho_X(x, x_0) < \delta \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon.$$

Step 1: Identify the parts that will change.

We will work from left to right through the bold parts. We want to describe when continuity fails at this same x_0.

▸ A numerical example ▸ At a point versus on X ◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Negating Continuity Step by Step

Definition

A function $f : X \to Y$ is **not continuous** at x_0 if **there exists** $\epsilon_0 > 0$ such that the following statement is false:

There exists $\delta > 0$ such that, **for every $x \in X$**,

$$\rho_X(x, x_0) < \delta \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon_0.$$

Step 2: Change "for every" to "there exists."

Continuity must work for every allowed error. To make it fail, one allowed error is enough. Call it ϵ_0; for this value, the remaining statement must be false.

▸ A numerical example ▸ At a point versus on X ◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Negating Continuity Step by Step

Definition

A function $f : X \to Y$ is **not continuous** at x_0 if **there exists** $\epsilon_0 > 0$ such that, **for every** $\delta > 0$, the following statement is false:

For every $x \in X$,

$$\rho_X(x, x_0) < \delta \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon_0.$$

Step 3: Change "there exists" to "for every."

Saying that no δ works means that every $\delta > 0$ fails. For each one, the statement about all points must be false.

▸ A numerical example ▸ At a point versus on X ◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Negating Continuity Step by Step

Definition

A function $f : X \to Y$ is **not continuous** at x_0 if **there exists $\epsilon_0 > 0$** such that, **for every $\delta > 0$, there exists $x \in X$** for which the following implication is false:

$$\rho_X(x, x_0) < \delta \quad \Longrightarrow \quad \rho_Y(f(x), f(x_0)) < \epsilon_0.$$

Step 4: Change "for every point" to "there exists a point."
If the implication does not hold for every point, there must be at least one point where it fails. This point may depend on δ.

▸ A numerical example ▸ At a point versus on X ◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Negating Continuity Step by Step

Definition

A function $f : X \to Y$ is **not continuous** at x_0 if **there exists** $\epsilon_0 > 0$ such that, **for every** $\delta > 0$, **there exists** $x \in X$ with

$$\rho_X(x, x_0) < \delta$$
and it is not true that
$$\rho_Y(f(x), f(x_0)) < \epsilon_0.$$

Step 5: Make the implication false.

The input must satisfy $\rho_X(x, x_0) < \delta$, while the output must fail to satisfy $\rho_Y(f(x), f(x_0)) < \epsilon_0$. This is why the input inequality stays the same.

▸ A numerical example ▸ At a point versus on X ◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Negating Continuity Step by Step

Definition

A function $f : X \to Y$ is **not continuous** at x_0 if **there exists $\epsilon_0 > 0$** such that, **for every $\delta > 0$, there exists $x \in X$** with

$$\rho_X(x, x_0) < \delta \quad \textbf{and} \quad \rho_Y(f(x), f(x_0)) \geq \epsilon_0.$$

Step 6: Replace "not smaller than" with "greater than or equal to."

This gives the definition of discontinuity. The same ϵ_0 works for every δ; the point x may change.

▸ Try it: Negate a statement ▸ At a point versus on X ◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Exercise — Negating Continuity on X

Definition

A function $f : X \to Y$ is continuous on X if, for every point $x \in X$, f is continuous at x.

Exercise: Use this definition to state what it means for f to be not continuous on X. Identify the parts to change, then negate them one at a time.

◂ Continuity at a point ▸ Cheat sheet ▸ A numerical example

Appendix: Exercise — Negating Continuity on X

Definition

A function $f : X \to Y$ is **continuous** on X if, **for every point $x \in X$**, f is **continuous** at x.

Step 1: Identify the parts that will change.

The bold parts describe continuity on X, the requirement about every point, and continuity at the chosen point.

◂ Continuity at a point ▸ Cheat sheet ▸ A numerical example

Appendix: Exercise — Negating Continuity on X

Definition

A function $f : X \to Y$ is **not continuous** on X if **the following statement is false**:

For every point $x \in X$, f is **continuous** at x.

Step 2: Say that continuity on X fails.

Change *continuous on X* to *not continuous on X*. The defining statement about all points must now be false.

◂ Continuity at a point ▸ Cheat sheet ▸ A numerical example

Appendix: Exercise — Negating Continuity on X

Definition

A function $f : X \to Y$ is **not continuous** on X if **there exists at least one point $x \in X$** for which **the following statement is false**:

f is **continuous** at x.

Step 3: Change for every to there exists.

A property fails to hold at every point when there is at least one point where it fails. Keep the domain X unchanged.

◂ Continuity at a point ▸ Cheat sheet ▸ A numerical example

Appendix: Exercise — Negating Continuity on X

Definition

A function $f : X \to Y$ is **not continuous** on X if **there exists at least one point $x \in X$** at which f is **not continuous**.

Step 4: Negate continuity at that point.

Saying that f *is continuous at* x is false means that f is *not continuous at* x.

◂ Continuity at a point ▸ Cheat sheet ▸ A numerical example

Appendix: Exercise — Negating Continuity on X

Definition

A function $f : X \to Y$ is **not continuous** on X if **there exists at least one point $x \in X$** at which f is **discontinuous**.

Step 5: Use the word discontinuous.

Not continuous at x and *discontinuous at x* mean the same thing. This is just a change of wording.

◂ Continuity at a point ▸ Cheat sheet ▸ A numerical example

Appendix: Exercise — Negating Continuity on X

Definition

A function $f : X \to Y$ is **not continuous** on X if **there exists at least one point $x \in X$** at which f is **discontinuous**.

Step 6: Read and check the completed definition.

One point of discontinuity is enough. The function does not have to be discontinuous at every point of X.

◂ Continuity at a point ▸ Cheat sheet ▸ A numerical example

Appendix: The Same Error Defeats Every Radius

Consider $h : \mathbb{R} \to \mathbb{R}$ at $x_0 = 0$, where

$$h(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases} \qquad h(0) = 0.$$

Choose the allowed output error $\epsilon_0 = 1/2$ and keep it fixed.

Appendix: The Same Error Defeats Every Radius

Consider $h : \mathbb{R} \to \mathbb{R}$ at $x_0 = 0$, where

$$h(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases} \qquad h(0) = 0.$$

Choose the allowed output error $\epsilon_0 = 1/2$ and keep it fixed.

Proposed radius δ	A point inside: $x = \delta/2$	Output distance
0.1	0.05	$\lvert h(x) - h(0) \rvert = 1$
0.01	0.005	$\lvert h(x) - h(0) \rvert = 1$
0.001	0.0005	$\lvert h(x) - h(0) \rvert = 1$

Appendix: The Same Error Defeats Every Radius

Consider $h : \mathbb{R} \to \mathbb{R}$ at $x_0 = 0$, where

$$h(x) = \begin{cases} 0, & x \le 0, \\ 1, & x > 0. \end{cases} \qquad h(0) = 0.$$

Choose the allowed output error $\epsilon_0 = 1/2$ and keep it fixed.

Proposed radius δ	A point inside: $x = \delta/2$	Output distance
0.1	0.05	$\lvert h(x) - h(0)\rvert = 1$
0.01	0.005	$\lvert h(x) - h(0)\rvert = 1$
0.001	0.0005	$\lvert h(x) - h(0)\rvert = 1$

These examples suggest a rule that works for **every** $\delta > 0$: choose $x = \delta/2$. Then

$$|x - 0| = \delta/2 < \delta, \qquad |h(x) - h(0)| = 1 \ge 1/2.$$

No radius guarantees the chosen accuracy, so h is not continuous at 0.

◂ The definition ◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Failure of Continuity on X

1. A function f is continuous on X if it is continuous at **every** point $x_0 \in X$.

One point of discontinuity suffices.

◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Failure of Continuity on X

1. A function f is continuous on X if it is continuous at **every** point $x_0 \in X$.
2. So f is not continuous on X if there exists **at least one** point $x_0 \in X$ at which it is not continuous.

One point of discontinuity suffices.

◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: Failure of Continuity on X

1. A function f is continuous on X if it is continuous at **every** point $x_0 \in X$.
2. So f is not continuous on X if there exists **at least one** point $x_0 \in X$ at which it is not continuous.
3. Substitute the definition we just derived:
   There exists a point $x_0 \in X$ and a number $\epsilon_0 > 0$ such that, for every $\delta > 0$, there exists a point $x \in X$ with

$$\boxed{\rho_X(x, x_0) < \delta \quad \text{and} \quad \rho_Y(f(x), f(x_0)) \geq \epsilon_0.}$$

One point of discontinuity suffices.

◂ Exercise 4.5 ◂ Exercise 4.6

Appendix: From Discontinuity to a Sequence

Goal

If f is not continuous at x_0, show that there exists a sequence (x^n) converging to x_0 for which $f(x^n)$ does not converge to $f(x_0)$.

1. Take the fixed $\epsilon_0 > 0$ from negated continuity.

The image sequence may converge to a different value.

▸ Negating continuity ◂ Exercise 4.6

Appendix: From Discontinuity to a Sequence

Goal

If f is not continuous at x_0, show that there exists a sequence (x^n) converging to x_0 for which $f(x^n)$ does not converge to $f(x_0)$.

1. Take the fixed $\epsilon_0 > 0$ from negated continuity.
2. For each $n \geq 1$, set $\delta^n = 1/n$. Choose $x^n \in X$ such that

$$\rho_X(x^n, x_0) < 1/n, \qquad \rho_Y(f(x^n), f(x_0)) \geq \epsilon_0.$$

The image sequence may converge to a different value.

▸ Negating continuity ◂ Exercise 4.6

Appendix: From Discontinuity to a Sequence

Goal

If f is not continuous at x_0, show that there exists a sequence (x^n) converging to x_0 for which $f(x^n)$ does not converge to $f(x_0)$.

1. Take the fixed $\epsilon_0 > 0$ from negated continuity.
2. For each $n \geq 1$, set $\delta^n = 1/n$. Choose $x^n \in X$ such that

$$\rho_X(x^n, x_0) < 1/n, \qquad \rho_Y(f(x^n), f(x_0)) \geq \epsilon_0.$$

3. The first inequality gives $x^n \to x_0$. The second prevents $f(x^n) \to f(x_0)$: no tail lies inside the ϵ_0-ball around $f(x_0)$.

The image sequence may converge to a different value.

▸ Negating continuity ◂ Exercise 4.6

Appendix: Characterizations of Continuity

Let $f : X \to Y$, where X, Y are metric spaces.

$$
\begin{aligned}
& f \text{ continuous} \\
\iff\ & f^{-1}(V) \text{ open in } X \text{ for every open } V \subseteq Y \\
\iff\ & f^{-1}(C) \text{ closed in } X \text{ for every closed } C \subseteq Y.
\end{aligned}
$$

Definitions

Open: $U \subseteq X$ is open in X if for every $x \in U$ there is $r > 0$ with $B_X(x, r) \subseteq U$, where

$$B_X(x, r) = \{z \in X : \rho_X(z, x) < r\}.$$

Closed: $C \subseteq X$ is closed in X if $X \setminus C$ is open in X.

▸ Open-set proof ▸ Closed-set proof ◂ Exercise 5.2

Appendix: Open Sets — First Direction

$$\boxed{f \text{ continuous, } V \subseteq Y \text{ open } \implies f^{-1}(V) \text{ open in } X}$$

1. Take any arbitrary $x_0 \in f^{-1}(V)$. Since $f(x_0) \in V$ and V is open, choose $\epsilon > 0$ with

$$B_Y(f(x_0), \epsilon) \subseteq V.$$

▸ Second direction ◂ Exercise 5.2: Definitions

Appendix: Open Sets — First Direction

$$\boxed{f \text{ continuous}, \ V \subseteq Y \text{ open} \implies f^{-1}(V) \text{ open in } X}$$

1. Take any arbitrary $x_0 \in f^{-1}(V)$. Since $f(x_0) \in V$ and V is open, choose $\epsilon > 0$ with

$$B_Y(f(x_0), \epsilon) \subseteq V.$$

2. By continuity at x_0, choose $\delta_\epsilon > 0$ such that

$$\rho_X(x, x_0) < \delta_\epsilon \implies \rho_Y(f(x), f(x_0)) < \epsilon.$$

▸ Second direction ◂ Exercise 5.2: Definitions

Appendix: Open Sets — First Direction

$$\boxed{f \text{ continuous}, \ V \subseteq Y \text{ open} \implies f^{-1}(V) \text{ open in } X}$$

1. Take any arbitrary $x_0 \in f^{-1}(V)$. Since $f(x_0) \in V$ and V is open, choose $\epsilon > 0$ with

$$B_Y(f(x_0), \epsilon) \subseteq V.$$

2. By continuity at x_0, choose $\delta_\epsilon > 0$ such that

$$\rho_X(x, x_0) < \delta_\epsilon \implies \rho_Y(f(x), f(x_0)) < \epsilon.$$

3. Hence $B_X(x_0, \delta_\epsilon) \subseteq f^{-1}(V)$. Every point has a ball inside the preimage, so it is open. The empty preimage is open as well.

▸ Second direction ◂ Exercise 5.2: Definitions

Appendix: Open Sets — Second Direction

$$\boxed{\begin{array}{c} f^{-1}(V) \text{ open for every open } V \subseteq Y \\ \Longrightarrow f \text{ continuous} \end{array}}$$

1. Fix $x_0 \in X$ and any $\epsilon > 0$. The ball $V = B_Y(f(x_0), \epsilon)$ is open in Y.

▸ Closed-set proof ◂ Exercise 5.2: Second direction

Appendix: Open Sets — Second Direction

$$f^{-1}(V) \text{ open for every open } V \subseteq Y \implies f \text{ continuous}$$

1. Fix $x_0 \in X$ and any $\epsilon > 0$. The ball $V = B_Y(f(x_0), \epsilon)$ is open in Y.
2. Its preimage $U = f^{-1}(V)$ is open and contains x_0. Choose $\delta > 0$ such that $B_X(x_0, \delta) \subseteq U$.

▸ Closed-set proof ◂ Exercise 5.2: Second direction

Appendix: Open Sets — Second Direction

$$\boxed{\begin{array}{c} f^{-1}(V) \text{ open for every open } V \subseteq Y \\ \Longrightarrow f \text{ continuous} \end{array}}$$

1. Fix $x_0 \in X$ and any $\epsilon > 0$. The ball $V = B_Y(f(x_0), \epsilon)$ is open in Y.
2. Its preimage $U = f^{-1}(V)$ is open and contains x_0. Choose $\delta > 0$ such that $B_X(x_0, \delta) \subseteq U$.
3. Verify:

$$\rho_X(x, x_0) < \delta \implies f(x) \in V \implies \rho_Y(f(x), f(x_0)) < \epsilon.$$

▸ Closed-set proof ◂ Exercise 5.2: Second direction

Appendix: Open Sets — Second Direction

$$\boxed{\begin{array}{c} f^{-1}(V) \text{ open for every open } V \subseteq Y \\ \Longrightarrow f \text{ continuous} \end{array}}$$

1. Fix $x_0 \in X$ and any $\epsilon > 0$. The ball $V = B_Y(f(x_0), \epsilon)$ is open in Y.
2. Its preimage $U = f^{-1}(V)$ is open and contains x_0. Choose $\delta > 0$ such that $B_X(x_0, \delta) \subseteq U$.
3. Verify:

$$\rho_X(x, x_0) < \delta \implies f(x) \in V \implies \rho_Y(f(x), f(x_0)) < \epsilon.$$

For $Y = \mathbb{R}$, checking open intervals suffices: every open set is a union of open intervals, and $f^{-1}(\bigcup_i I_i) = \bigcup_i f^{-1}(I_i)$.

▸ Closed-set proof ◂ Exercise 5.2: Second direction

Appendix: Closed-Set Characterization

$$\boxed{\begin{array}{c} f \text{ continuous} \iff f^{-1}(C) \text{ closed in } X \\ \text{for every closed } C \subseteq Y \end{array}}$$

Use complements and the open-set characterization:

$$f^{-1}(Y \setminus A) = X \setminus f^{-1}(A).$$

1. **First direction:** Take any closed $C \subseteq Y$. Then $Y \setminus C$ is open. Continuity implies that $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ is open. Hence $f^{-1}(C)$ is closed.

Appendix: Closed-Set Characterization

$$\boxed{\begin{array}{c} f \text{ continuous} \iff f^{-1}(C) \text{ closed in } X \\ \text{for every closed } C \subseteq Y \end{array}}$$

Use complements and the open-set characterization:

$$f^{-1}(Y \setminus A) = X \setminus f^{-1}(A).$$

1. **First direction:** Take any closed $C \subseteq Y$. Then $Y \setminus C$ is open. Continuity implies that $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ is open. Hence $f^{-1}(C)$ is closed.
2. **Second direction:** Assume all closed sets have closed preimages. For any open $V \subseteq Y$, the set $Y \setminus V$ is closed. Its preimage is closed, so its complement $f^{-1}(V)$ is open. Thus f is continuous.

▸ Characterizations ◂ Definitions ◂ Exercise 5.2: First direction

Appendix: Definition of Convergence

Definitions

Convergent: A sequence (x^n) converges to x if, for every $\varepsilon > 0$, there exists N such that $\rho(x^n, x) < \varepsilon$ for all $n \geq N$.

◂ Exercise 4.6: First direction